



Machine Learning Prediction of On/Off Target-driven Clinical Adverse Events

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Abstract

Objective Currently, 90% of clinical drug development fails, where 30% of these failures are due to clinical toxicity. The current extensive animal toxicity studies are not predictive of clinical adverse events (AEs) at clinical doses, while current computation models only consider very few factors with limited success in clinical toxicity prediction. We aimed to address these issues by developing a machine learning (ML) model to directly predict clinical AEs.

Methods Using a dataset with 759 FDA-approved drugs with known AEs, we first adapted the ConPLex ML model to predict IC₅₀ values of these FDA-approved drugs against their on-target and off-target binding among 477 protein targets. Subsequently, we constructed a new ML model to predict clinical AEs using IC₅₀ values of 759 drugs' primary on-target and off-target effects along with tissue-specific protein expression profiles.

Results The adapted ConPLex model predicted drug-target interactions for both on- and off-target effects, as shown by co-localization of the 6 small molecule kinase inhibitors with their respective kinases. The coupled ML models demonstrated good predictive capability of clinical AEs, with accuracy over 75%.

Conclusions Our approach provides a new insight into the mechanistic understanding of *in vivo* drug toxicity in relationship with drug on/off-target interactions. The coupled ML models, once validated with larger datasets, may offer advantages to directly predict clinical AEs using *in vitro/ex vivo* and preclinical data, which will help to reduce drug development failure due to clinical toxicity.

Keywords On-target · Off-target · Machine learning · Adverse events · Drug tissue distribution

Introduction

Currently, 90% of clinical drug development fails, where 30% of these failures are due to clinical toxicity. [1–4]. This results in the high developmental cost of drugs that make

it into the market. Even after being approved, many drugs are withdrawn from the market for showing high incidents of AEs. The clinical toxicity poses a significant challenge to both pharmaceutical industries and health care [5, 6], therefore, there is a critical need to accurately and efficiently predict the safety of new drug candidates in humans.

To reduce the risk of clinical toxicity, the current investigational new drug (IND) application for initiation of a clinical phase I study requires extensive GLP (good laboratory practice) toxicity studies in three different animal species. [7] Although the GLP toxicity studies can identify and eliminate compounds with severe toxicity, they rarely predict clinical AEs of the compound at clinically relevant therapeutic dose ranges [8]. An analysis of 2,366 drugs concluded that “results from tests on animals are highly inconsistent predictors of toxic responses in humans”. [9] This inconsistency has prompted Congress to enact the “21st Century FDA Modernization Act”, which encourages the use of alternative approaches, such as *in vitro* screening and computation methods, to replace animal toxicity testing [10].

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Recently, Machine learning (ML) models have been applied to predict drug toxicity. However, several challenges remain.

First, a majority of ML models have focused on predicting *in vitro* cellular toxicity from chemical structures with various methodologies for molecular representations. [4] For instance, the prediction of cellular activity against nuclear receptors is used as the endpoint in the Tox21 Challenge, a subset of the broader “Toxicology in the 21st Century” initiative. [11] The Tox21 challenge provides prediction from 12 *in vitro* assays that test seven different nuclear receptor signaling effects and five stress response effects of 10,000 molecules in cells. However, *in vitro* toxicity prediction may not necessarily be in concordance with clinical AEs in humans, thus reducing their ability to predict clinical toxicity.

The second challenge of ML models is to determine which parameters/features to be used in the ML model in predicting toxicity. Since ML models have increasingly shifted towards deep learning models, which are “black-box” models with limited explainability, it is essential to determine which parameters need to be built into the models to gain confidence in toxicity prediction. [12] Some efforts in explaining toxicity predictions have focused on “toxicophores”, a chemical structure or a functional group, that is related to the toxic properties. [13] Quantitative structure-activity relationship (QSAR) models were used to computationally predict drug promiscuity and toxicity. [14] These models correlate specific “toxicophores” present in each compound’s structure with the *in vitro* toxicity. Although these models allow a good correlative interpretation of the predictions, these methods generally do not examine the effect of the absence of these structure features in the drug. Additionally, it is not clear if these features are adequate/sufficient for accurate prediction of *in vivo* clinical toxicity.

The third challenge to use ML to predict toxicity is to understand the drug-target interactions *in vitro* and *in vivo*. Most ML models assume that toxicity is only caused by drugs’ on-targets in the major organs and thus building the ML models using known drug-target interactions as input features. However, it is known that drugs’ *in vivo* effective concentration at clinically relevant dose is usually much higher than the *in vitro* IC₅₀ values of the drugs binding to the on-targets. Therefore, the clinical AEs of the drugs at clinically relevant doses will not only be driven by interactions with on-targets but also with unknown off-targets. [15]

Therefore, we hypothesize clinical AEs of drugs are influenced by the following aspects in this study: (1) structure (*S*) information of the drug; (2) potency/specificity (*P/S*) of drug binding to the on- or off-targets in the major organs at relevant clinical concentrations in human; and (3) the expression levels (*E*) of these on-targets and off-targets in the major organs.

In this study, we used the collection of 759 FDA-approved drugs, for which clinical AEs data were available. Firstly, we used advanced methodologies to represent the drug structures using molecular fingerprints. Then, we adapted ConPLex [16] ML model to predict drugs’ potency/specificity for the drug-target interactions with on-targets with IC₅₀ < 30 nM (i.e., on-target), and drug-target interactions with off-targets with IC₅₀ of 30-250 nM (i.e., off-target). Finally, we developed a novel ML model to predict the clinical AEs using these predicted on- and off-targets, while taking into account the expression profiles of these on- and off-targets across various tissues Fig. 1.

Methods

Data

Adverse Events (AEs). The frequency data pertaining to drug AEs was obtained from the study conducted by Galeano et al [17]. Subsequently, this data was transformed into a binary format, where a value of -100 represents missing values (removed from further ML model building), a value of 0 represents infrequent occurrences, and a value of 1 denotes frequent occurrences of the respective side effects. The dataset encompasses information on 759 distinct drugs, with a specific focus on the prevalence of the four most common AEs: namely, nausea, diarrhoea, dizziness, and headache.

Molecular Similarity. In order to gain a deeper understanding of the chemical composition diversity of the included 759 molecules within our dataset, we conducted a comprehensive analysis. This involved calculating the Tanimoto similarity index of compounds, utilizing Molecular ACCess Systems (MACCS) keys fingerprint as the input, as described by Bajusz et al [18]. To further demonstrate the diversity of our dataset, we also used Morgan fingerprint with length 2048 and radius 2 for the Tanimoto similarity analysis.

In addition to calculating the Tanimoto similarity, we conducted Principle Component Analysis (PCA) using the MACCS fingerprint as our chosen method for representing drugs in our dataset. It is important to acknowledge that PCA has its limitations, particularly when dealing with data that possesses inherent non-linearity in its features. Recognizing this constraint, we also employed an alternative technique, t-distributed stochastic neighbor embedding (t-SNE), to address the need for more effective data clustering and the visualization of drug properties.

Protein Expression Level Information. Target expression levels across different tissues and cell types were extracted from the Human Protein Atlas [19]. Data is available for a

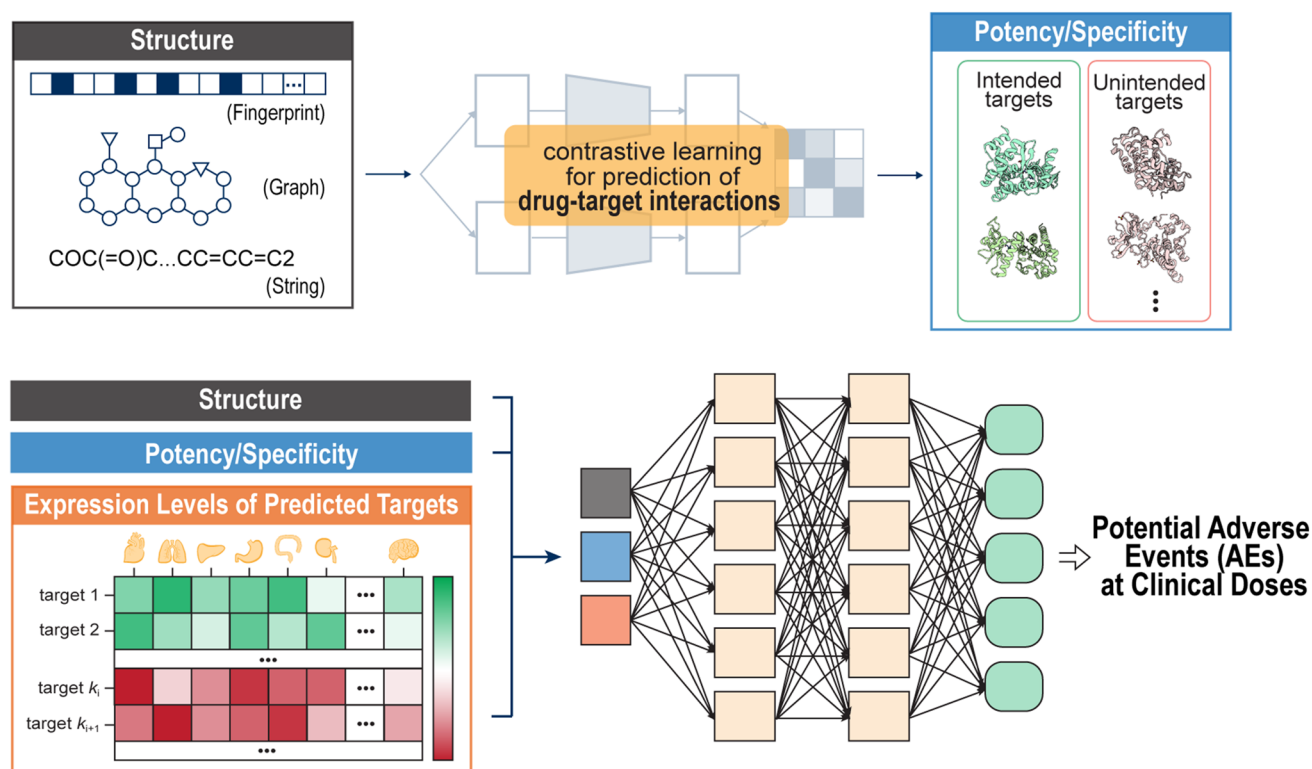


Fig. 1 The workflow of data integration and ML modeling.

total of 15,004 distinct genes, encompassing various tissues throughout the human body. Expression level is represented as High, Medium, Low, and Not detected, as well as Not representative.

Drug Potency/specificity. The IC_{50} values represent the compound/substance concentrations required for 50% inhibition of the targets. The pIC_{50} values are defined as $-\log(IC_{50})$. If available, the IC_{50} or pIC_{50} for different targets were extracted from Therapeutic Data Commons (TDC) [20] and BindingDB [21]. ConPLex was adapted to predict IC_{50} or pIC_{50} values for drugs to 477 protein targets if the IC_{50} information cannot be found in public data sets [16]. Details about ConPLex are shown in the later section.

Tissue/Cell Selectivity. We posit that a drug's selectivity for a tissue/cell type is intricately linked to the IC_{50} values concerning targets situated within that specific tissue. Essentially, higher affinities towards those targets have the potential to effectively "pull" compounds into the tissue. This estimation process involves multiplying the pIC_{50} values by the expression levels of targets present within specific cell types within the target tissue. The resulting products are then aggregated to represent the drug's selectivity within the designated tissue as shown in Eq. 1:

$$S_{D,C} = \sum_i A_{D,T_i} \cdot E_{T_i,C} \quad (1)$$

in which $S_{D,C}$ denotes the "Selectivity" of a drug in a certain tissue cell type, where D is the drug and C is the tissue cell type. $A_{D,T}$ represents pIC_{50} value for a drug D and target T , and $E_{T,C}$ denotes the expression level of a target in a cell type.

Model Architecture

Adapting ConPLex for pIC_{50} Prediction [16]. Protein target features were generated using ProtBert and the length n protein embedding $E_{full} \in \mathbb{R}^{n \times d_t}$ was mean-pooled along the protein length to obtain vector $T \in \mathbb{R}^{d_t}$ to finally represent the protein. Drug $M \in \mathbb{R}^{d_m}$ was featured by Morgan fingerprint [22]. A single fully connected layer architecture was used to convert protein and drug embedding into embeddings of the same dimension, $T^*, M^* \in \mathbb{R}^h$. The probability $\hat{p}(T^*, M^*)$ of drug-target interaction was calculated as:

$$\hat{p}(T^*, M^*) = \sigma\left(\frac{T^* \cdot M^*}{\|T^*\|_2 \cdot \|M^*\|_2}\right) \quad (2)$$

For predicting affinity value $\hat{y}(T^*, M^*)$, the calculation was:

$$\hat{y}(T^*, M^*) = \text{ReLU}(T^*, M^*) \quad (3)$$

Binary cross-entropy loss was used for probability prediction and mean squared error loss was used for binding affinity value prediction.

Contrastive learning was used to improve prediction performance. A training set of triplets (T, M^+, M^-) was collected, in which T represents the protein target, positive samples M^+ denote known interacting molecules toward target T , and negative samples M^- represent the decoys against target T . The aim of contrastive learning is to minimize the distance between T and the positive compounds M^+ while maximizing the distance between T and the decoys M^- . Distance D could be computed as $1 - \hat{p}$, $\hat{p}$ representing the probability of drug-target interaction predicted by the model. Compounds and targets were mapped into latent space embedding of dimension $\mathbb{R}^h$ and the aim was to minimize triplet margin-distance loss L_{TRM} :

$$L_{\text{TRM}} = \frac{1}{N} \sum_{i=1}^N \max(D(a, p) - D(a, n) + m, 0) \quad (4)$$

in which m is the margin.

Clustered Heat Maps for Visualizing pIC₅₀ Predictions. An interactive clustered heat map builder (available at <https://build.ngchm.net/NGCHM-web-builder/>) was utilized to present the predicted pIC₅₀ values of each drug against each binding target. The dataset is organized with drugs in rows while targets in columns. The default settings of iterative steps were adopted, including the heat map format, where color blue is associated with lower threshold, color white is value 0, and color red is associated with higher threshold. In other words, red color implies greater pIC₅₀ values, i.e., stronger binding, as compared to white or blue color. Hierarchical clustering was also performed using Ward's algorithm with Euclidean distance metric for both variables (i.e., drugs and targets).

Multilayer Perceptron (MLP) for AE Prediction. MLP was used to predict adverse event frequency given the input of drug "Selectivity" as defined above. The architecture for MLP with K layers is demonstrated below:

$$h^{(1)} = \text{ReLU}(W^{(1)} \cdot x) \quad (5)$$

$$h^{(2)} = \text{ReLU}(W^{(2)} \cdot h^{(1)}) \quad (6)$$

$$\vdots \quad (7)$$

$$h^{(K)} = \text{Sigmoid}(W^{(K)} \cdot h^{(K-1)}) \quad (8)$$

in which x represents a vector containing the calculated drug "Selectivity" in each tissue/cell type. We used the ReLU activation function for each neural layer, and the Sigmoid activation function was used to convert the predicted values into probabilities.

Weighted binary cross entropy loss Eq. 9 was used as the loss function given the imbalance of labels inside the data set:

$$\mathcal{L}(y, \hat{y}) = y \log \hat{y} + w(1 - y) \log(1 - \hat{y}) \quad (9)$$

in which $\hat{y}$ is the predicted probability and y denotes the true AE label, and weight w regulates the penalty associated with incorrect predictions for a particular class.

Performance Metrics. The equations for calculating each metric are provided below, with TP representing true positives, TN for true negatives, FP for false positives, and FN for false negatives.

$$\text{accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (10)$$

$$\text{precision} = \frac{TP}{TP + FP} \quad (11)$$

$$\text{recall (SE)} = \frac{TP}{TP + FN} \quad (12)$$

$$\text{F1 score} = \frac{2 \cdot (\text{precision} \cdot \text{recall})}{\text{precision} + \text{recall}} \quad (13)$$

Model Training and Implementation. The dataset was divided into training set, validation set, and testing set with a ratio of 0.68:0.17:0.15. The random state seed was set to be 42 to avoid potential information leak from test set. Weights of the model were updated with loss backpropagation using the AdamW optimizer with learning rate initialization of 10^{-5} and weight decay of 10^{-5} . Early stopping was employed to address issues related to overfitting and accelerate the training process. In the event that there was no improvement in performance on the validation set for a continuous span of 30 epochs, the training process would be halted. The results on test set were used to evaluate the model performance. Model was implemented in PyTorch. Model training and inference were performed using CPU or a single NVIDIA A100 GPU.

Results

Data Analysis

The heat map of molecular similarity plotted by the Tanimoto similarity index using MACCS fingerprint and Morgan fingerprint of 759 drugs as input is visually presented in Fig. 2a and b. The data showed that the similarity index for all molecules in our dataset tends to be relatively low. This observation suggests that the molecules contained within our dataset showcase a broad spectrum of structural diversity. This is important because it implies that our dataset serves as an excellent representation of the chemical space. More-

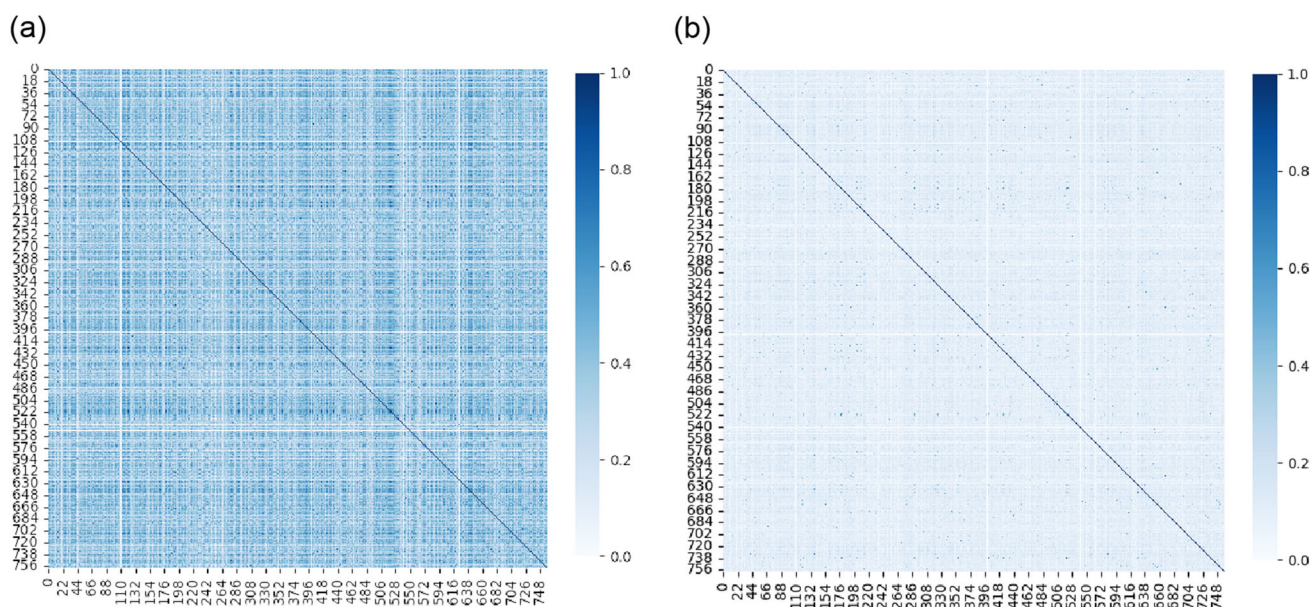


Fig. 2 Heat maps of molecular similarity plotted by the Tanimoto similarity index using (a) MACCS and (b) Morgan fingerprint of the 759 drugs.

over, it underscores the versatility of our model in predicting AEs for compounds possessing a wide range of structural variations.

As illustrated in Fig. 3, the outcome of the t-SNE visualization reveals that compounds display overlapping distributions regardless of their labels. A comparable overlap is evident in the PCA analysis (Figure S1), suggesting that neither PCA nor t-SNE can effectively cluster compounds into groups based on their labels. This observation suggests that within the vast landscape of chemical space, distinguishing between compounds labeled for different AEs is not a straightforward task. It also indicates that certain compounds exhibit significant structural and property similarities, which contribute to this overlapping pattern. The results highlight the complexity of predicting AEs based solely on chemical structure and property information, emphasizing the multifaceted nature of drug behavior and the need for more advanced ML modeling approaches to capture these intricate relationships.

Adapting ConPLex ML to Predict pIC_{50} of on-/off-targets of 759 FDA Approved Drugs

The ConPLex ML model was adapted to predict the binding affinity (represented as pIC_{50}) to the on- and off-targets of 759 FDA-approved drugs. We first used ConPLex ML model to predict the pIC_{50} values of all 759 drugs against 477 possible targets (Fig. 4). It is apparent that drugs have significantly different binding profiles. The drugs shown on the top

half of the heat map have stronger binding affinities across the panel of selected 477 potential binding targets, however, the strength of binding varies for different targets, as evidenced by lighter red in the top left session as compared to darker red in the top right session. The drugs in the bottom half generally have weaker binding across the selected targets as labeled with blue or light red, but it is noted that some of those drugs may also have strong binding for targets in the center. This observation implies the importance of considering both on- and off-targets for predicting drug toxicities, given the complex biological system inside human.

We further studied the ML prediction for the IC_{50} against the primary targets ($\text{IC}_{50} < 30$ nM), and primary/secondary (or unintended) targets ($\text{IC}_{50} < 250$ nM) of 6 FDA-approved small molecule kinase inhibitors (SMKIs) in our database (i.e., afatinib, lapatinib, gefitinib, erlotinib, ruxolitinib, and tofacitinib). We only included those targets that at least one SMKI drug binds/inhibits with $\text{IC}_{50} < 30$ nM (total 198 targets) or $\text{IC}_{50} < 250$ nM (total 308 targets). As presented in (Fig. 5), heat map clustered epidermal growth factor receptor (EGFR) inhibitors (i.e., afatinib, lapatinib, gefitinib, and erlotinib) into a group as they share similar binding patterns across all targets. Similarly, two Janus kinase (JAK) inhibitors (i.e., ruxolitinib and tofacitinib) were grouped together based on similar binding patterns. The correct clustering or co-localization of kinases and corresponding kinase inhibitors demonstrates the state-of-the-art performance of ConPLex in predicting drug-target interactions. As shown in Table 1, the total number of primary targets for these 6 drugs ranges from 28 to 73 (on average 33 targets per drug). In com-

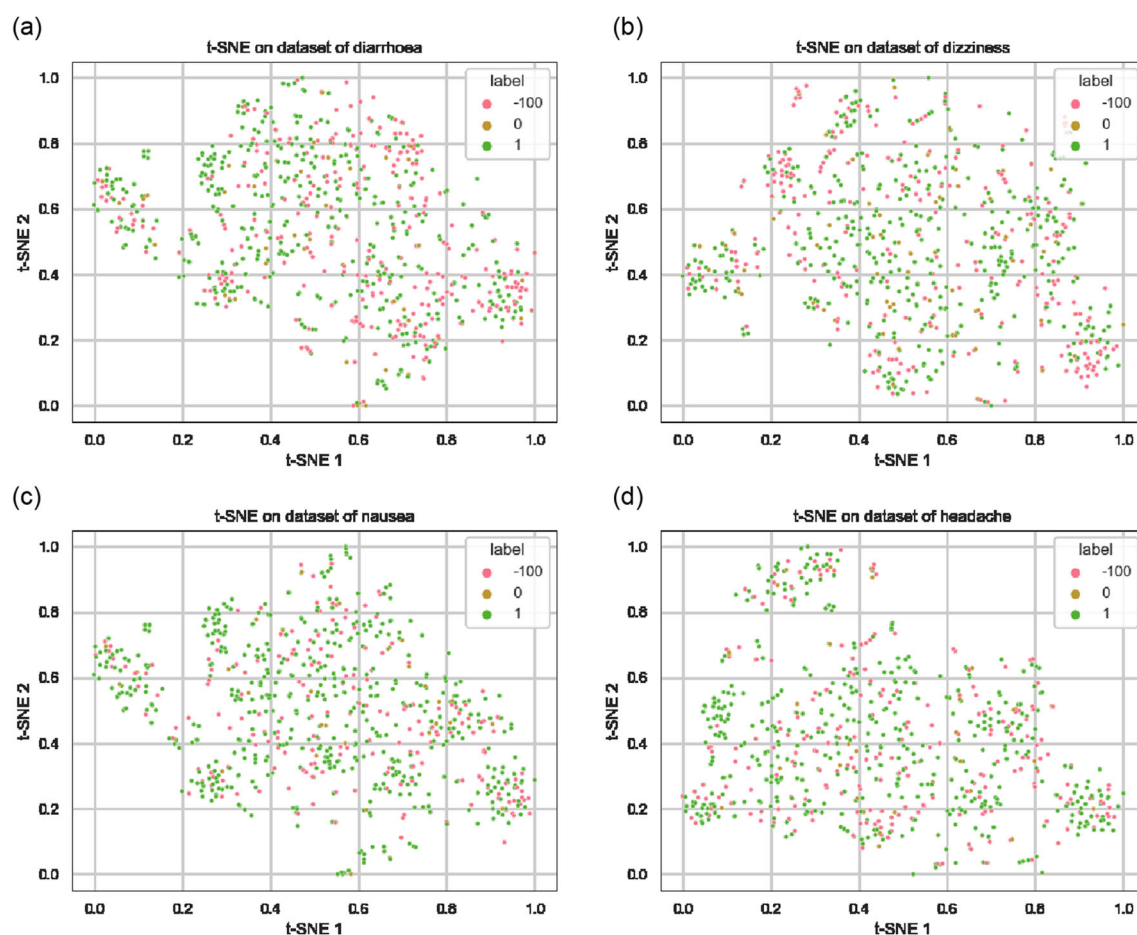


Fig. 3 t-SNE visualizations of drugs, labeled based on the frequency of the (a) diarrhoea, (b) dizziness, (c) nausea, and (d) headache.

Fig. 4 Heatmap of predicted pIC_{50} values of 759 FDA-approved drugs against 477 targets (X-axis represents targets while Y-axis represents drugs)

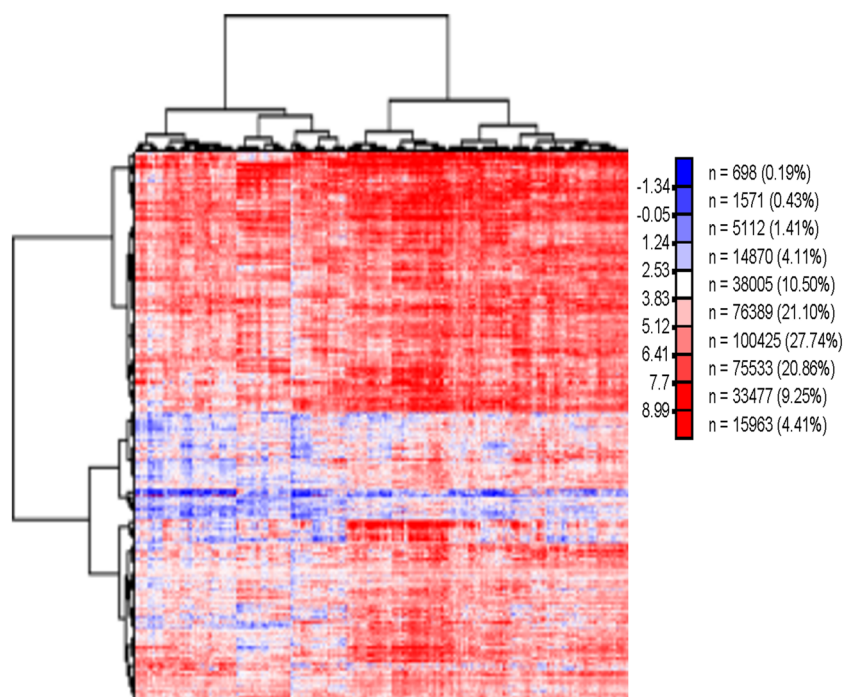
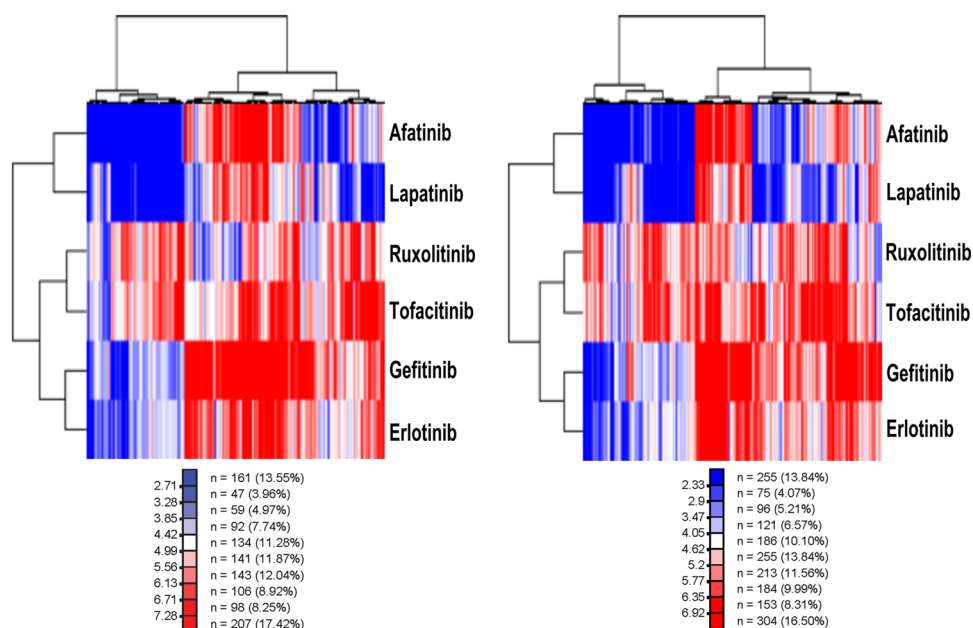


Fig. 5 Heatmaps of predicted pIC_{50} values of 6 FDA-approved SMKI drugs' primary targets (left) and primary/unintended targets (right)



parison, the total number of primary and off-targets for these 6 drugs ranges from 51 to 120 (on average 66 primary/off-targets per drug).

ML model to Predict Clinical AEs from on- and off-targets

We used the ML model to predict the clinical AEs using the above predicted pIC_{50} values against on- and off-targets, while taking into account the expression profiles of these on/off-targets across various tissues. It is worth noting that a drug's structure would direct the drug's binding potential with different targets, therefore, only drugs' potency/specificity values and expression levels of different targets were included in the ML model as inputs, to avoid co-linearity. The performance of our model in predicting various AEs, as summarized in Table 2, exhibited consistent and good overall predictive performance across the spectrum of evaluated AEs. In predicting headache, nausea, diarrhoea, and dizziness, the model showed overall good performance with accuracy, precision, recall, and F1 scores

each exceeding the thresholds of 0.86, 0.88, 0.96, and 0.92, respectively. Detailed confusion matrices illustrating the prediction performance for each AE are provided in Figure S2. We also further calculated Cohen's kappa score, which reflects the level of agreement between the prediction and true labels. [23] The Cohen's kappa statistic values are 0.411 for headache, 0.413 for nausea, 0.419 for diarrhoea, and 0.311 for dizziness. This suggests that the model successfully classifies cases involving these AEs. It is worth noting that our model is limited by the small and imbalanced dataset that needs to be further improved using larger datasets, but it provides a proof of concept for prediction of AEs based on on/off-targets.

Discussion

Leveraging multiple levels of domain knowledge in drug characteristics and system pharmacology in building this ML model seeks to not only improve prediction accuracy but also provide deeper insights into the underlying mechanisms of

Table 1 Total number of binding targets for 6 SMKIs

Drug	$IC_{50} < 30$ nM (n=198 targets)	$IC_{50} < 250$ nM (n=308 targets)
afatinib	28	51
lapatinib	6	21
ruxolitinib	12	46
tofacitinib	37	97
gefitinib	73	120
erlotinib	42	58
Average Number/Drug	33	66

Table 2 Performance of the ML model for predicting four different AEs

AE Prediction	accuracy	precision	recall	F1
Headache	0.936	0.959	0.973	0.966
Nausea	0.940	0.963	0.975	0.969
Diarrhoea	0.930	0.928	1.000	0.962
Dizziness	0.865	0.884	0.968	0.924

drug toxicity. We intended to incorporate different aspects to directly predict clinical AEs: (1) the potency/specificity (P/S) of drug binding to the on- and off-targets in the major organs at relevant clinical concentrations in human; and (2) the expression levels of these on-targets and off-targets in the major organs. Our ML model sheds light on the role of on-target and off-target effects of drug-target interactions in causing clinical AEs. To our best knowledge, this research is innovative and is the first ML model that combines ML methods with complex biological networks in using drug on-/off-targets interaction to predict clinical toxicity.

Advanced ML methodologies developed for drug toxicity prediction typically base their approaches on molecular chemical structures, or merge this with biological properties of the molecule through multi-modal methods. Such multi-modal techniques have demonstrated enhanced performance by utilizing curated molecule qualities, consisting of target proteins, enzymes, pathways, and phenotypic indications. Nevertheless, merely factoring in the chemical and biological properties overlooks the intricate *in vivo* interactions between drugs and potential protein targets (i.e., on- and off-targets) across various tissues. These interactions can significantly influence the drug's *in vivo* distribution and biological effect, thereby exerting its therapeutic effectiveness and potential toxicity.

Indeed, multiple researches have been conducted to investigate these overlooked drug-off-target interactions during drug development, which suggests that the off-targets are indeed contributing to the adverse effects of drugs. [24–26] For example, Lin et al. discovered that seven of the seven clinical cancer drug candidates they tested were confirmed to fail to inhibit their intended targets and exhibited off-target toxicities. [27] Additionally, the Computational Analysis of Novel Drug Opportunities (CANDO) has been used to create a comprehensive drug-protein interaction signature that can predict a drug's therapeutic potential based on its interaction on a proteomic scale. [28, 29] However, these models did not include the expression of the off-targets in different tissues for AE prediction. The AEs caused by these off-targets are caused by both the binding affinity of a drug to its targets

and the expression profile of these proteins in the organs. We hypothesize that the drug's retention in specific tissues is determined by the expression level of target proteins and the binding affinity, thereby causing more pharmacological side effects.

To mitigate this gap, we proposed an interaction scheme between drugs and various on-/off-targets in different tissues. This scheme affects the *in vivo* distribution of drugs (as distinct targets manifest differential expression levels in various tissues) and ultimately their action sites, resulting in drug toxicity. In this initial exploration, we designed an ML model incorporating protein expression variables in a tissue-specific context, which demonstrated an overall accuracy in AE frequency prediction. Further, our model indicated a correlation between the predicted instances of side effects and the high expression level of interactive proteins, regardless of whether these proteins were classified as intended pharmacological targets or unintended off-targets.

Some potential limitations to our current model are also noted. In this initial effort, while we incorporated ~477 proteins [20], spanning a broad spectrum of protein types as targets, in the training of the regression-based drug-target interaction prediction, these proteins only represent a minimal fraction of the proteins in the drug-targets in the human genome (estimated to be 30,000 proteins) [3]. This deficient profile of target proteins could potentially impact the predictive performance. Moreover, our dataset of 759 drug molecules is still small. Future work needs to be expanded to include more drug molecules from the FDA-approved drug bank. Lastly, tissue/cell selectivity needs to be experimentally measured in animal models to estimate that in human organs. Altogether, our work attempts to build an ML model that accounts for the neglected interactions between drug and off-targets in a tissue-specific manner in predicting clinical toxicity and further exploration and rigorous model validations are warranted.

Conclusion

Currently, 90% of clinical drug development fails, where 30% of these failures are due to clinical toxicity. The current extensive animal toxicity studies are not predictive of clinical AEs at clinical doses, while current computation models only consider very few factors with limited success in clinical toxicity prediction. We hypothesize that ML models, integrating a drug's potency/specificity to the on- and off-drug targets, driven by drug structures, and target expression levels in different tissues/organs, could help to predict clinical drug toxicity.

We first applied the ConPLex ML model to predict pIC₅₀ values of 759 FDA-approved drugs interacting with their on- and off-targets among 477 proteins. Using a dataset of 759 FDA drugs, we constructed an ML model to predict clinical AEs using the predicted potency/specificity by pIC₅₀ values of these 759 drugs against on-/off-targets, and tissue-specific protein interaction profiles in varied tissues. The resulting ML model demonstrated a predictive capability of clinical AEs such as headache and nausea. Our approach provides insights into the mechanistic understanding of *in vivo* clinical AEs in relationship to drug-on/off-target interaction. The ML models, once validated with large datasets, may offer advantages to directly predict clinical AEs using *in vitro/ex vivo*, preclinical data, which help to reduce drug development failure due to clinical toxicity.

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Declarations

Conflicts of interest The authors declare no conflict of interest.

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